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## ORTHOPEDIC STAPLE AND RELATED INSTRUMENTS

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### Abstract

Novel orthopedic staples and related instruments are disclosed. An instrument is an orthopedic drill guide assembly that includes a first arm and a second arm that are retractably connected to the distal end of the instrument's outer housing, where the proximal ends of the arms are retractably connected to the distal end of the outer housing and configured to move in a retractable motion that changes the spacing between the first drill guide sleeve and the second drill guide sleeve, where the proximal end of the outer housing is configured for controlling the retractable motion of the first and second arms.

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## **Background/Summary**

### **CROSS-REFERENCE TO RELATED APPLICATION**

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference in accordance with 37 C.F.R. §§ 1.57; 1.97; and 1.98.

### **FIELD OF DISCLOSURE**

[0002] The present disclosure relates generally to orthopedic staples.

### **BACKGROUND**

[0003] Orthopedic staples are utilized for moving and/or maintaining bones or bone parts substantially adjacent to one another after certain surgical procedures. Moreover, orthopedic staples are often utilized to compress and hold together in place two displaced pieces of bone while healing. Often a hole or holes are necessary in the bone material for proper insertion of the surgical staples.

### **SUMMARY**

[0004] Disclosed herein are novel orthopedic staples and related instruments.

[0005] Provided is an orthopedic drill guide assembly comprising: an elongated body having a distal end, a proximal end, and a longitudinal vertical plane defined therethrough, wherein the elongated body comprising: an outer housing having a proximal end and a distal end; and a connecting piece provided within the outer housing; a first arm and a second arm retractably connected to the distal end of the outer housing, wherein each of the first arm and the second arm comprises a distal end and a proximal end; a first drill guide sleeve located at the distal end of the first arm; and a second drill guide sleeve located at the distal end of the second arm; wherein the first drill guide sleeve and the second drill guide sleeve are separated by a spacing, wherein the proximal ends of the arms are retractably connected to the distal end of the outer housing and configured to move in a retractable motion that changes the spacing between the first drill guide sleeve and the second drill guide sleeve, wherein the proximal end of the outer housing is configured for controlling the retractable motion of the first and second arms.

[0006] Also provided is an orthopedic drill guide assembly comprising: an elongated body having a distal end and a proximal end, and a longitudinal vertical plane defined therethrough; a first guide hole and a second guide hole located near the distal end of the elongated body, wherein the first and second guide holes are arranged in a linear alignment with the longitudinal vertical plane of the elongated body such that the first guide hole is between the second guide hole and the proximal end of the elongated body; a distal sliding portion slidably engaging the distal end of the elongated body and is configured to slide back and forth along a direction that is in-line with the longitudinal vertical plane of the elongated body, wherein the distal sliding portion is communicably connected to the proximal end of the elongated body and the distal sliding portion's sliding motion

is actuated and controlled from the proximal end of the elongated body; and a third guide hole located on the distal sliding portion and in-line with the first and second guide holes along the longitudinal vertical plane of the elongated body, wherein the first and second guide holes are located between the third guide hole and the proximal end, whereby the distal sliding portion's sliding motion changes the distance between the third guide hole and the first guide hole.

[0007] Also provided is an orthopedic drill guide assembly comprising: an elongated body having a distal end and a proximal end, and a longitudinal vertical plane defined therethrough; a first drill guide hole and a second drill guide hole located near the distal end of the elongated body, wherein the first and second drill guide holes are arranged in a linear alignment with the longitudinal vertical plane of the elongated body such that the first drill guide hole is between the second drill guide hole and the proximal end of the elongated body

[0008] Also provided is a bone staple comprising: a first leg; a second leg; and a bridge connecting the two legs, the bridge comprising a top surface along the length of the bridge, wherein the top surface comprises: two chamfered edge surfaces that extend substantially the length of the bridge; and one flat surface between the two chamfered edge surfaces, wherein the flat surface extends substantially the length of the bridge, wherein each of the first end and the second end of the bridge transitions down to a leg.

[0009] Also provided is a bone staple comprising: a pair of legs, each having a proximal end and a distal end, the pair of legs defining a compression plane; and a non-linearly extending bridge portion connecting the distal ends of the pair of legs, wherein the bridge portion has a generally U-shape and comprises: a flat portion that spans the distance between the distal ends of the pair of legs; and two ends of the U-shape connecting the flat portion to the proximal ends of the legs, wherein the two ends of the U-shape extending laterally between the flat portion and the proximal ends of the legs, whereby the flat portion is off-set from the compression plane; wherein the flat portion has a width that is greater than its thickness.

[0010] Also provided is a surgical staple comprising: a bridge that has an open loop configuration and a thickness from 0.05 upto 0.1 inches; two or more pairs of legs extending from the bridge in the same direction, wherein each of the legs has a surface that faces the other leg in its pair, wherein the surface comprises a plurality of barbs.

[0011] Also provided is a surgical staple comprising: a bridge having an elongated shape with a longitudinal axis defined through its length, the bridge further comprising first, second, third, and fourth corners; a first leg extending from the first corner; a second leg extending from the second corner; a third leg extending from the third corner; a fourth leg extending from the fourth corner; the bridge comprising a first section that includes the first and second corners and a second section that includes the third and fourth corners; and wherein the first and second corners are arranged such that a line connecting the first and second corners intersect the longitudinal axis of the bridge at a first non-orthogonal angle.

[0012] Also provided is a bone staple inserter kit comprising: a first sterile package containing a first bone staple securely mounted to a first cartridge, wherein the first bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs, wherein the first cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge portion of the first bone staple and securely hold the first bone staple by an interference fit; and a second sterile package containing: a bone staple insertion tool that is configured to receive a cartridge, like the first cartridge that has a bone staple, like the first bone staple, securely mounted thereon; a universal drill guide assembly; and a drill bit, wherein the bone staple insertion tool comprises an actuation mechanism that can transform the bone staple to an insertion shape after the cartridge with the bone staple mounted thereon is mounted on the insertion tool.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The various embodiments of the inventive hydrogel implant of the present disclosure will be described in more detail in conjunction with the following drawing figures. The structures in the drawing figures are illustrated schematically and are not intended to show actual dimensions.

[0014] FIGS. 1A-2D illustrate an embodiment of an orthopedic drill guide assembly for predrilling holes in two bone pieces for orthopedic staples.

[0015] FIGS. 3A and 3B illustrate another embodiment of an orthopedic drill guide assembly for predrilling holes in two bone pieces for orthopedic staples.

[0016] FIGS. 4A to 4H illustrate another embodiment of an orthopedic drill guide assembly for predrilling holes in two bone pieces for orthopedic staples.

[0017] FIGS. 5A-5F are illustrations of shape memory metal staples having a chamfered bridge according to an embodiment.

[0018] FIGS. 6A and 6B illustrate a universal staple holder for inline bone staples according to an embodiment of the present disclosure.

[0019] FIGS. 7A to 7D illustrate the universal bone staple holder of FIGS. 6A, 6B interacting with a universal staple inserter for inline bone staples according to an embodiment of the present disclosure.

[0020] FIG. 8 is an illustration of an overlapping view of three different size staples that are configured for using with the universal staple holder shown in FIGS. 6A and 6B.

[0021] FIGS. 9A and 9B in combination show a surgical kit providing inline bone staples in the universal staple holders shown in FIG. 6A and the associated universal instruments: the orthopedic drill guide shown in FIG. 4A; and a universal staple inserter shown in FIG. 7A.

[0022] FIGS. 10A-10D are illustrations of shape memory metal staples having an offset bridge according to an embodiment.

[0023] FIGS. 11A-11C are illustrations showing a staple inserter for the offset bridge staples of FIGS. 10A-10D.

[0024] FIGS. 12A-12E are illustrations showing another shape memory metal staple according to another embodiment.

[0025] FIGS. 13A-13D are illustrations showing another shape memory metal staple according to another embodiment.

[0026] FIGS. 14A-14I are illustrations showing a universal staple inserter system of the present disclosure.

[0027] FIG. 15 is an illustration of a surgical kit according to the present disclosure.

### DETAILED DESCRIPTION

[0028] This description of the exemplary embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. The drawing figures are not necessarily to scale and certain features may be shown exaggerated in scale or in somewhat schematic form in the interest of clarity and conciseness. In the description, relative terms such as “horizontal,” “vertical,” “up,” “down,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and normally are not intended to require a particular orientation. Terms including “inwardly” versus “outwardly,” “longitudinal” versus “lateral” and the like are to be interpreted relative to one another or relative to an axis of elongation, or an axis or center of rotation, as appropriate. Terms concerning attachments, coupling and the like, such as “connected” and “interconnected,” refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable

or rigid attachments or relationships, unless expressly described otherwise. When only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. The term “operatively connected” is such an attachment, coupling or connection that allows the pertinent structures to operate as intended by virtue of that relationship. In the claims, means-plus-function clauses, if used, are intended to cover the structures described, suggested, or rendered obvious by the written description or drawings for performing the recited function, including not only structural equivalents but also equivalent structures.

[0029] Referring to FIGS. 1A-2, an embodiment of an orthopedic drill guide assembly **100** for predrilling holes in two bone pieces for orthopedic staples is disclosed. The orthopedic drill guide assembly **100** comprises an elongated body formed by a first housing **110** having a distal end **111** and a proximal end **112**, and a longitudinal vertical plane Pv defined therethrough. The first housing **110** comprises a handle portion **110A** and a neck portion **110B**. A first guide hole **121** and a second guide hole **122** are located near the distal end **111** of the elongated body. The first and second guide holes **121**, **122**, respectively, are arranged in a linear alignment with the longitudinal vertical plane Pv of the elongated body such that the first guide hole **121** is between the second guide hole **122** and the proximal end **112** of the elongated body.

[0030] The first and second guide holes **121**, **122** are the drill guides that are used to predrill holes into bone repair site where an orthopedic staple will be placed. Therefore, the distance between the first and second guide holes **121**, **122** is set to represent the base distance between the two legs of an orthopedic staple. The term “base distance” refers to the distance between the two legs of an orthopedic staple that is in its insertion configuration. In other words, the state where the two legs of the staple are parallel to each other and ready to be inserted into a bone repair site.

[0031] At the distal end **111** is provided a distal sliding portion **142** that slidably engages the distal end of the elongated body and is configured to slide back and forth along a direction that is in-line with the longitudinal vertical plane of the elongated body. The distal sliding portion **142** is communicatively connected to the proximal end **112** of the elongated body and the distal sliding portion's sliding motion is actuated and controlled from the proximal end **112** of the drill guide assembly **100**. The elongated body also comprises a third guide hole **123** located on the distal sliding portion **142** and in-line with the first **121** and second **122** guide holes along the longitudinal vertical plane Pv of the elongated body. The first and second guide holes **121**, **122** are located between the third guide hole **123** and the proximal end **112**, whereby the distal sliding portion's sliding motion changes the distance between the third guide hole **123** and the first guide hole **121**. The distal sliding portion **142** is part of a second housing **140** that connects the distal sliding portion **142**, and in turn the third guide hole **123**, to the proximal end **112** of the drill guide assembly **100**.

[0032] The orthopedic drill guide assembly **100** further comprises a knob **160** provided at the proximal end **112** of the elongated body. The knob **160** is communicatively connected to the distal sliding portion **142** by the second housing **140** whereby the sliding motion of the distal sliding portion **142** for changing the distance between the third guide hole **123** and the first guide hole **121** is actuated and controlled by turning the knob **160**. By turning the knob **160** in one direction, the second housing **140** along with its sliding portion **142** can be made to slide toward the proximal end **112** decreasing the distance between the first and third guide holes **121**, **123**. By turning the knob **160** in the opposite direction, the second housing **140** and the distal sliding portion **142** can be made to slide away from the proximal end **112** increasing the distance between the first and third guide holes **121**, **123**.

[0033] During use, the drill guide assembly **100** is positioned over a bone repair site (e.g., joint/osteotomy/fracture) so that the first and second guide holes **121**, **122** span over the repair site. In this position, the first guide hole **121** will be on one side of the repair site and the second and the

third guide holes **122**, **123** will be on the opposite side of the repair site. Then, a K-wire or a fixation pin (**P1**, **P2**) is placed through each of the first and third guide holes **121** and **123** then driven into the respective bone pieces on the two sides of the repair site. In the illustrated example shown in FIG. 2, one of the holes for the staple is first predrilled into one of the two bone pieces using the first guide hole **121** as a drill guide. Next, a first pin **P1** is inserted through the first guide hole **121** and into the predrilled hole in the first bone piece. Next, a second pin **P2** is inserted through the third guide hole **123** and into the second bone piece. Then, the knob **160** is turned to reduce the distance between the first and third guide holes **121**, **123** thereby compressing the repair site (joint/osteotomy/fracture) until the two bone pieces involved are in contact. After the desired compression is achieved, the repair site should be positioned between the first and the second guide holes **121** and **122**, so that the first guide hole **121** affixed to a first bone piece by the first pin **P1** and the second guide hole **122** is positioned over a second bone piece. Then, the second guide hole **122** is used as a drill guide and a hole is drilled into the second bone piece. Next, the second pin **P2** is removed and the drill guide **100** is removed. The surgeon can now manually compress the repair site and implant a staple into the predrilled holes. The use of the drill guide assembly **100** can be useful for predrilling holes for staples when the joint, osteotomy, or fracture have some gap between the two bone pieces. In those situations, if a shape memory metal staple is inserted without proper temporary fixation or compression of the two bone pieces, the staple will first bring the two bone pieces together and close the gap as the staple returns to its memorized shape. However, that also brings the legs of the staple inward reducing the amount of compression force the staple exerts on the repair site.

[0034] In some embodiments of the orthopedic drill guide assembly **100**, rather than the knob **160**, a lever can be provided at the proximal end **112** of the elongated body for actuating to pull in the third guide hole **123** toward the first guide hole **121** and reduce the distance between the third guide hole **123** and the first guide hole **121**. The lever would be communicatively connected to the distal sliding portion **142** by a connecting piece that sits within the elongated body **110**, whereby the sliding motion of the distal sliding portion **142** is actuated and controlled by moving the lever.

[0035] In some embodiments, the drill guide assembly **100** is configured so that the distance between the first and second guide holes **121**, **122** is adjustable so that the drill guide assembly **100** can be universally used for various different staple sizes. FIGS. 2B-D illustrate in more detail an example of the internal structure of the drill guide assembly **100** that enables the feature of adjusting the distance between the first and second guide holes **121**, **122**. FIG. 2B is a top view of the drill guide assembly **100**. FIG. 2C shows a longitudinal cross-sectional view of the drill guide assembly **100** taken through the section line B-B shown in FIG. 2B.

[0036] As shown in FIG. 2D, which is an exploded view of the drill guide assembly **100**, the elongated body portion comprises three primary components: the first housing **110**; the second housing **140**, and a core stem **130**. The second guide hole **122** is provided near the distal end **111** of the neck portion **110B** of the first housing **110**.

[0037] The second housing **140** comprises the distal sliding portion **142** near the distal end **111** as discussed above. The second housing **140** is formed of two elongated portions **140A**, **140B**. The two elongated portions are connected at the distal sliding portion **142** but are spaced apart and form a space **140C** therebetween. At the proximal end **112**, the two elongated portions **140A**, **140B** are provided with a threaded portion **144** for threadably engaging the knob **160**.

[0038] In the fully assembled drill guide assembly **100**, the neck portion **110B** of the first housing **110** is received in the space **140C** between the two elongated portions **140A**, **140B**. Thus, near the distal end **111** of the drill guide assembly, the distal sliding portion **142** wraps around the neck portion **110B** as shown in FIGS. 1A and 2A. The proximal half of the two elongated portions **140A**, **140B**, however, are positioned inside the handle portion **110A** of the first housing **110**. To accomplish this arrangement, the handle portion **110A** is only connected to the top surface portion of the neck portion **110B** and forms a spacing **110C** between the handle portion **110A** and the neck

portion **110B** as noted in FIG. 2D. The spacing **110C** is open at the bottom side so that the two elongated portions **140A**, **140B** can slip into the spacing **110C** from the bottom side. In this arrangement, the knob **160** rotatably engages the groove **110D** provided at the proximal end **112** of the first housing **110** via the knob's rim **161**. The threaded portion **144** of the second housing **140** is received into the interior **162** of the knob which is threaded with corresponding female threads. This arrangement can be better seen in the cross-sectional view in FIG. 2C. Thus, by turning the knob **160**, the threaded engagement enables moving the second housing **140** and the distal sliding portion **142** back and forth in distal-proximal direction described above.

[0039] The neck portion **110B** of the first housing **110** is open from the bottom side noted by the arrow E in FIG. 2D. In the fully assembled drill guide assembly **100**, the core stem **130** is slipped into the open space within the neck portion **110B**. The neck portion **110B** is provided with an opening **116** to accommodate the fixation pin P1 or a drill bit that will be inserted into the first guide hole **121**. The opening **116** is elongated to accommodate the full sliding motion of the core stem **130** for adjusting the spacing between the first guide hole **121** and the second guide hole **122**. The core stem **130** is provided with a slider button **132** that is positioned between the first guide hole **121** and the proximal end **112** of the drill guide assembly **100**.

[0040] The core stem **130** is also provided with a visual marker tab **135**. The handle portion **110A** can comprise an elongated opening **115** through which the slider button **132** protrudes to allow the user to push pull on the slider button **132** to move the core stem **130** back and forth to move the first guide hole **121** back and forth along a direction (distal-proximal) that is in-line with the longitudinal vertical plane Pv of the drill guide assembly **100**.

[0041] The handle portion **110A** can also comprise a plurality of slots **114** through which the visual marker tab **135** can at least partially protrude for indicating the position of the first guide hole **121** with respect to the second guide hole **122**. Because the first and second guide holes **121**, **122** are the drill guide holes that will be used to drill holes for the bone staples, this ability to change the spacing between the first and second guide holes **121**, **122** allow the drill guide assembly **100** to be used for many different sizes of staples.

[0042] Therefore, the plurality of slots **114** can be located at preset distances representing particular staple size that corresponds to the selected distance between the first and second guide holes **121**, **122**. Each of the plurality of slots **114** corresponds to a particular staple size and the visual marker tab **135** can be a colored marking that moves along with the slider button **132** so that the user can see the marker through the slots **114** and tell when a desired spacing between the first and second guide holes **121**, **122** has been selected corresponding to the desired staple size. In some embodiments, by having the visual marker tab **135** at least partially protrude through the slots **114**, the visual marker tab **135** can be made to click through the slots **114** thus providing additional tactile feedback to the user. As shown in FIG. 2D, the portion **136** of the core stem **130** on which the slider button **132** and the visual marker tab **135** reside can be configured to be a cantilevered structure to allow the slider button **132** and the visual marker tab **135** to be pressed down during the sliding motion. Additionally, when the visual marker tab **135** pops into a slot **114**, the visual marker tab can prevent unintended movement of the slider button **132** and lock drill guide assembly to the selected staple size.

[0043] In the drill guide assembly **100**, the second guide hole **122** is provided on the first housing **110** at a fixed location. The second guide hole **122** is referred to as being at a fixed location because it is the first guide hole **121** that moves with respect to the position of the second guide hole **122** in the frame of reference of the drill guide assembly **100**. By moving the first guide hole **121**, the distance between the first guide hole **121** and the second guide hole **122** can be adjusted.

[0044] The handle portion **110A** is shaped to make it comfortable for the user to grab and hold the drill guide **100**. The slider button **132** can be provided in the handle portion so that the user can manipulate the slider button **132** with the user's thumb for one-handed operation.

[0045] FIG. 1B illustrates two different top-down views of the slots **114**, the slider button **132**, and

the visual marker tab **135**, each view illustrating two different staple size selection. In the view shown on the left side, the slider button **132** is in its most distal position (closest to the distal end **111** of the drill guide assembly. In this example, when the slider button **132** is in this position, the first guide hole **121** is at its closest to the second guide hole **122**. Thus, this position represents selection for the smallest staple size. In the view shown on the right side of FIG. **1B**, the slider button **132** is in its most proximal position (closest to the proximal end **112** of the drill guide assembly. In this example, when the slider button **132** is in this position, the first guide hole **121** is at its furthest from the second guide hole **122**. Thus, this position represents selection for the largest staple size.

[0046] Referring to FIGS. **3A-3B**, according to another aspect, a simpler drill guide assembly **100A** is disclosed. The drill guide assembly **200** is similar in construction as the drill guide assembly **100A** but the drill guide assembly **200** is not configured to perform the pre-drilling temporary compression of the repair site. The drill guide assembly **100A** does not have the third guide hole **123** of the drill guide assembly **100**. The drill guide assembly **100A** comprises the same first and second guide holes **121**, **122**, the slider button **132** and the plurality of slots **114** as in the drill guide assembly **100**. Thus, drill guide assembly **100A** is useful for predrilling holes in two bone pieces in a repair site for a staple.

[0047] Referring to FIGS. **4A-4H**, an orthopedic drill guide assembly **300** according to another embodiment is disclosed. The drill guide assembly **300** comprises an elongated body **305** having a distal end **311** and a proximal end **312**, and a longitudinal vertical plane Pv defined therethrough. The elongated body **305** comprises an outer housing **310** having a proximal end and a distal end, and a connecting piece **370** provided within the outer housing **310**. The drill guide assembly **300** also comprises a first arm **351** and a second arm **352** retractably connected to the distal end **311** of the outer housing **310**. The first arm **351** comprises a distal end **351A** and a proximal end **351B**. The second arm **352** comprises a distal end **352A** and a proximal end **352B**.

[0048] The distal ends **351A**, **352A** of the first and second arms **351**, **352**, respectively, are provided with a first drill guide sleeve **321** and a second drill guide sleeve **322**, respectively. The first drill guide sleeve **321** and the second drill guide sleeve **322** are separated by a spacing D (See FIG. **4B**).

[0049] The proximal ends **351B**, **352B** of the arms are retractably connected to the distal end **311** of the outer housing **310** and configured to move in a retractable motion that changes the spacing D between the first drill guide sleeve **321** and the second drill guide sleeve **322**.

[0050] The drill guide assembly **300** further comprises a knob **350** provided at the proximal end **312** of the outer housing **310** for actuating and controlling the retractable motion of the first and second arms **351**, **352**. The knob **350** is communicatively connected to the first and second arms **351**, **352**.

[0051] FIG. **4C** shows a longitudinal cross-section of the drill guide assembly **300** where the section is taken through the line A-A shown in FIG. **4B**. The connecting piece **370** has a proximal end **372** and a distal end **371** and extends through the outer housing **310** and provides the communicating connection between the knob **350** and the first and second arms **351**, **352**.

[0052] The knob **350** is connected to the proximal end **372** of the connecting piece **370** and the connection between the knob **350** and the connecting piece **370** is configured to move the connecting piece **370** back and forth between the distal end **311** and the proximal end **312** of the outer housing **310**.

[0053] The proximal ends **351B**, **352B** of the first and second arms **351**, **352**, respectively, are pivotally connected to a pivot point **380** located at the distal end **371** of the connecting piece **370**. The pivot point **380** is on the longitudinal vertical plane Pv. The pivotal connection at the pivot point **380** is made by a pivot pin **381**.

[0054] Referring to FIG. **4F**, each of the first and second arms **351**, **352** are configured with an elongated slot **355**, **356**, respectively, that extends between the arm's proximal end **351B**, **352B** and

the distal end **351A**, **352A**. The first and second arms **351**, **352** are arranged so that the elongated slot **355**, **356** on each of the first and second arms **351**, **352** is oriented outward at an acute angle  $\alpha$  from the longitudinal vertical plane Pv of the elongated body **305**. The first and second arms **351**, **352** overlap one another at their proximal ends **351B**, **352B** where the pivotal connection is made and a portion of the elongated slots **355**, **356** overlap one another. The distal end **311** of the outer housing **310** comprises a fixed pin **315** that extends through both of the elongated slots **355**, **356** at the overlapping portion, wherein the fixed pin **315** is on the longitudinal vertical plane Pv. As noted above, the pivot point **380** and the fixed pin **315** are both on the longitudinal vertical plane Pv. The fixed pin **315** keeps the two arms **351**, **352** from freely pivoting about the pivot point **380** without control. This arrangement allows the user to control the retractable motion of the first and second arms **351**, **352** by moving the connecting piece **370** back and forth using the knob **350**. The back and forth motion of the connecting piece **370** produces the retractable motion of the first and second arms **351**, **352** shown by the arrows **390** in FIG. 4B. The retractable motion of the first and second arms **351**, **352** changes the spacing D between the first drill guide sleeve **321** and the second drill guide sleeve **322**.

[0055] FIGS. 4B and 4D are illustrations of the drill guide assembly **300** with its first and second arms **351**, **352** in their fully extended configuration. The connecting piece **370** has been pushed toward the distal end **311** of the outer housing **310** as far as it can go. In turn, the connecting piece **370** has pushed the first and second arms **351**, **352** to their fully extended position because the first and second arms **351**, **352** are connected to the connecting piece **370** by the pivot pin **381**. As can be seen, each arm has been pushed out to its maximum extension and the fixed pin **315** is at the most proximal ends of the two slots **355**, **356**. In this configuration, the spacing D between the first and second arms is at its maximum.

[0056] In some embodiments of the drill guide **300**, the knob **350** and the connecting piece **370** are connected by screw thread arrangement that enables the back and forth movement of the connecting piece **370** by turning the knob **350**. In the illustrated example shown in FIG. 4C and 4G, the proximal end **372** of the connecting piece **370** is configured with a male screw thread **374**. As shown in FIGS. 4C and 4H, the internal bore of the knob **350** is provided with a female screw thread **354** that threadedly engage the proximal end **372** of the connecting piece **370**.

[0057] In some embodiments of the drill guide **300**, the knob **350** can be configured to have a clicking feature as the knob is rotated. Such clicking feature provides both tactile and aural feedback to the user so that the user knows where each position is when the knob is turned to the preset locations correlating to the sizes of the staples. The knob **350** can be provided with grooves **350A** as shown in FIG. 4H that interact with a protrusion **350B** (See FIG. 4C) on the proximal portion of the outer housing **310**.

[0058] In some embodiments of the drill guide **300**, each of the first and second drill guide sleeves **321**, **322** are provided with a hole sized to receive a drill bit. As shown in FIGS. 4A and 4E, in some embodiments of the drill guide, each of the first and second drill guide sleeves **321**, **322** extend parallel to each other.

[0059] Preferably, the connecting piece **370** moves back and forth along the longitudinal vertical plane Pv.

[0060] Referring to FIGS. 5A-5E, according to another aspect of the present disclosure, a surgical bone staple **1000** having a chamfered bridge portion is disclosed. The surgical bone staple **1000** comprises a first leg **1011**, a second leg **1012**, and a bridge **1020** connecting the two legs. The bridge **1020** has a top surface **1022** along the length of the bridge between a first end and a second end. The top surface **1022** comprises two chamfered edge surfaces **1023** and **1025** that extend substantially the length of the bridge **1020** and one flat surface **1024** between the two chamfered edge surfaces **1023**, **1025**. The flat surface **1024** also extends substantially the length of the bridge **1020**. As shown in the end-view of the staple **1000** in FIG. 5C, the bridge **1020** portion is wider than the width of the first and second legs **1011**, **1012**. This makes the bridge **1020** stronger and

more resistant to unwanted bending in the lateral plane. In this context the lateral plane refers to the plane that is substantially orthogonal to the legs.

[0061] In some embodiments, each of the two legs **1011**, **1012** comprises an inner surface **1017** and a plurality of barbs or teeth **1015** configured to prevent backing out of the staple after installation in a bone. Each of the plurality of barbs **1015** are formed by removing material between each barb such that the pointed tips of the barbs do not protrude beyond the inner surface of the legs. This is shown in FIGS. 5B and 5E by the dotted line S that represents the inner surface of the legs.

[0062] Each of the two legs **1011**, **1012** comprises a proximal end PE and a distal end DE and each distal end includes a wedge-shaped tip. In this context, the term “proximal” refers to the end of a staple leg that is attached to the bridge **1020** and the term “distal” refers to the end of a staple leg that is away from the bridge **1020**. Each of the two ends of the bridge transitions down to a leg. The transition from the bridge **1020** to the legs along the top surface **1024** of the bridge has a first radius of curvature R1 and the transition from the bridge to the legs along the bottom surface of the bridge has a second radius of curvature R2, wherein  $R1 > R2$ . Preferably,  $R1 = 0.110$  inches and  $R2 = 0.100$  inches. In some embodiments, the bone staple **1000** is comprised of a shape memory material such that the staple is movable between an insertion shape and a relaxed shape.

[0063] FIGS. 5D and 5E are illustrations of a bone staple **1000A** that has all the features of the bone staple **1000** discussed above but is a smaller sized staple. According to another embodiment, the bone staple **1000** can have more than two legs. FIG. 5F shows a bone staple **1000B** that is an example of such variation. The bone staple **1000B** comprises four legs **1011**, **1012**, **1013**, **1014**. Two pairs of legs **1011**, **1013** and **1012**, **1014** are provided. The first pair of legs **1011**, **1013** is on one side of the staple **1000B** and the second pair of legs **1012**, **1014** is on the other side of the staple **1000B**. In some embodiments, each of the legs can have different lengths as desired.

[0064] Referring to FIGS. 10A-10D, according to another aspect, a bone staple **2000** having an offset bridge portion is disclosed. The bone staple **2000** comprises a pair of legs **2011**, **2012**, each having a proximal end PE and a distal end DE. The pair of legs **2011**, **2012** define a compression plane CP. The bone staple **2000** also comprises a non-linearly extending (i.e., offset) bridge portion **2020** connecting the proximal ends of the pair of legs. The bridge portion **2020** has a generally U-shape and comprises a flat portion **2024** that spans the distance between the proximal ends of the pair of legs. Two ends of the U-shape connect the flat portion **2024** to the proximal ends PE of the legs **2011**, **2012**. The two ends of the U-shape extend laterally between the flat portion **2024** and the proximal ends PE of the legs **2011**, **2012**. The flat portion **2024** is offset from the compression plane CP. The flat portion **2024** has a width that is greater than its thickness to provide enhanced lateral stiffness. The ratio of the width/thickness of the flat portion **2024** can be about 2.0 to 2.5.

[0065] In some embodiments of the bone staple **2000**, the legs **2011**, **2012** comprise inner surfaces **2014** that are facing one another and a plurality of barbs **2015** along the inner surfaces **2014**. In some embodiments, the bone staple **2000** is comprised of a shape memory material such that the staple is movable between an insertion shape and an implanted shape.

[0066] FIG. 10C is a side view of the bone staple **2000**. FIG. 10D is an illustration showing an example of three different size offset bridge bone staples in a nested arrangement. This illustration of the nested arrangement shows that the bone staple embodiment **2000** of different sizes can be used to place multiple staples in close proximity to each other in the patient.

[0067] Referring to FIGS. 12A-12E, a bone staple **3000** according to another embodiment is disclosed. Two or more pairs of legs **3011**, **3012**, **3013**, **3014** extend from the bridge **3020** all in the same direction. Each of the legs **3011**, **3012**, **3013**, **3014** has a surface that faces the other leg in its pair, where the surface comprises a plurality of barbs **3015**. In the illustrated example, the bone staple **3000** is shown with two pairs of legs: **3011** and **3012** being one pair and **3013** and **3014** being the other pair. As show, each pair of legs have a plurality of barbs **3015** on the interior surface that faces the other leg in the pair. The bone staple **3000** is comprised of a shape memory material such that the staple is movable between an insertion shape (shown in FIG. 12B) and a

relaxed shape (shown in FIG. 12D).

[0068] The bone staple **3000** is not an intermedullary device and the bridge **3020** has an open-loop configuration and has a thickness T of about 0.05-0.1 inches. This thickness of the bridge **3020** allows easy removal of the staple from the repair site because it is easy to grab.

[0069] The configuration of the staple **3000** shown in FIG. 12A is its as-fabricated shape. For example, the staple **3000** can be machined out of a shape memory metal tubing (e.g. Nitinol) and retain the circular cross-sectional shape of the tube. FIG. 12C is the top down view showing the circular outline of the open-loop bridge **3020** as-fabricated from a tube.

[0070] FIG. 12D shows the outline of the open-loop bridge **3020** that has been compressed into an elongated oval-like shape and heat set to set the memory metal's memorized shape. The locations of the legs **3011**, **3012**, **3013**, and **3014** are shown by the dotted lines. The spacing between the legs of each pair of legs is set to d1.

[0071] FIGS. 12B and 12E show the insertion shape of the open-loop bridge **3020** after it has been cold worked into a shape to be held by a staple holder/insertor. The configuration shown in FIGS. 12B and 12E is achieved by bending the two sections of the bridge **3020**, one section between the first pair of legs **3011** and **3012**, and another section between the second pair of legs **3013** and **3014**, straight. The result is that the legs in each pair are moved further apart to a larger spacing d2. This configuration is the shape of the bone staple **3000** to be held by an inserting tool for implantation. Four holes are first drilled into the bone repair site using a specific drill guide for that purpose. Then, the four legs of the bone staple **3000** in the insertion shape shown in FIGS. 12B, 12E is inserted into those holes. As the bone staple **3000** reaches the activation temperature in the patient's body, the bridge **3020** of the bone staple **3000** returns to the memory set shape (i.e., the relaxed shape) shown in FIG. 12D. During this transformation, the two straight segments of the staple **3000** will move outward as indicated by the arrows E in FIG. 12E. This causes the other two segments of the bridge **3020** (the segments on top and bottom of FIG. 12E to move inward toward each other as indicated by the arrows C. This will generate compression force in the same direction as the arrows C as the legs try to return to the relaxed shape of FIG. 12D.

[0072] Referring to FIGS. 13A-13D, a bone staple **4000** having four staple legs in offset arrangements according to some other embodiments are disclosed. In these figures, the bone staple **4000** is shown in its insertion shape, as the staple legs are all shown in parallel arrangement. The bone staple **4000** comprises a bridge **4020** having an elongated shape with a longitudinal axis L defined through its length. The bridge **4020** further comprises a first corner **4021**, a second corner **4022**, a third corner **4023**, and a fourth corner **4024**. The bone staple **4000** also comprises a first leg **4011** extending from the first corner **4021**, a second leg **4012** extending from the second corner **4022**, a third leg **4013** extending from the third corner **4023**, and a fourth leg **4014** extending from the fourth corner **4024**. As shown, the first leg **4011** and the third leg **4013** are configured to be the pair providing the necessary compressive force when the staple is implanted into a bone repair site. The second leg **4012** and the fourth leg **4014** are configured to be the second pair providing the necessary compressive force when the staple is implanted into a bone repair site. As shown in FIG. 13A, the two legs that are paired for providing the compression function have a plurality of barbs **4015** on the surface that face each other. The barbs **4015** are oriented to prevent the staple from backing out once the staple is placed into a bone.

[0073] The bridge **4020** can also comprise a central reinforcement ridge **4027** that extends down the length of the bridge **4020**. The central ridge **4027** is thicker than the rest of the bridge and reinforces the strength and rigidity of the bridge **4020**. When the thickness of the central ridge **4027** is appropriately thick, the rigidity of the bridge **4020** can be comparable to a bone plate. In some embodiments, the central ridge **4027** is the thickest near the center of the bridge **4020** and tapers to the flat portion of the bridge **4020** near the two ends of the bridge **4020** to provide a smooth profile.

[0074] The bridge **4020** comprises a first section that includes the first and second corners **4021**,

**4022** and a second section that includes the third and fourth corners **4023, 4024**. The first and second corners **4021, 4022** are arranged such that a line **L1** connecting the first and second corners **4021, 4022** intersect the longitudinal axis **L** of the bridge **4020** at a first non-orthogonal angle  $\theta_1$ . [0075] In some embodiments of the bone staple **4000**, the third and fourth corners **4023, 4024** are arranged such that a line **L2** connecting the third and fourth corners **4023, 4024** intersect the longitudinal axis of the bridge at a second non-orthogonal angle  $\theta_2$ . In some embodiments, as shown in FIG. **13A**, the first non-orthogonal angle  $\theta_1$  and the second non-orthogonal angle  $\theta_2$  are different. Thus, the bone staple **4000** is configured so that the first pair of staple legs **4011, 4013** have greater spacing between them compared to the second pair of staple legs **4012, 4014**. The first pair of staple legs **4011, 4013** can span across a greater distance in the bone compared to the second pair of staple legs **4012, 4014**. An example of use for such bone staple **4000** is for metatarsophalangeal (MTP) fusion. Having four offset legs, the bone staple **4000** can be much more robust in being useful in various situations depending on the particular arrangement and physiology of the patient. The bone staple **4000** can be fabricated to have different offset configuration to fit the needs of different situation.

[0076] In some embodiments, the first non-orthogonal angle  $\theta_1$  and the second non-orthogonal angle  $\theta_2$  can be the same angle. An example of such configuration is shown in the bone staple **4000A** shown in FIG. **13D**. In this embodiment, the first pair of staple legs **4011A, 4013A** and the second pair of staple legs **4012A, 4014A** have the same spacing between the legs in each pair. The two pairs are just offset from each other and parallel to the longitudinal axis **L** of the bridge portion **4020**. Thus, the four corners **4021A, 4022A, 4023A, 4024A** and their respective staple legs **4011A, 4012A, 4013A, 4014A** each form a parallelogram arrangement as shown in FIG. **13D**.

[0077] As shown in the embodiment of the bone staple in FIG. **13D**, the line **L1** connecting the first and second corners **4021A, 4022A** and the line **L2** connecting the third and fourth corners **4023A, 4024A** can be parallel.

[0078] According to some embodiments, each of the four legs on the staple **4000, 4000A** can have a length that is independent of one another as can be seen in FIG. **13B** to accommodate particular anatomical locations. For example, the bone staple **4000** shown in FIG. **13B** where the two legs **4011** and **4012**, which are one leg in each of the two pairs, are longer than their paired legs **4013** and **4014**, respectively, can be useful where one side of the bone repair/susion site has a larger bone piece than the other. The longer legs would go into the larger bone piece.

[0079] In some embodiments, all four of the staple legs on the bone staple **4000, 4000A** can be of the same length.

[0080] In some embodiments, the bridge **4020** has an upper surface that curves down ward from the center line **L** to the four corners **4021, 4022, 4023, 4024**.

[0081] According to an aspect of the present disclosure, FIGS. **6A-7D** show a bone staple insertion system. The bone staple insertion system comprises a staple holding cartridge **200** and a universal staple inserter **400**. FIGS. **6A-6B** show a cartridge **200** configured for holding a two-legged inline bone staple such as the staple **1000** (shown in FIG. **5A**) that facilitates staple insertion procedure.

[0082] The cartridge **200** comprises an outer frame **210** and an inner frame **220**. The inner frame **220** is configured to releasably hold a bone staple **1000** in its insertion shape. In the insertion shape, the bridge portion **1020** of the staple **1000** is straightened with the two legs **1011** and **1012** opened so that they are substantially parallel to each other.

[0083] The inner frame **220** comprises a base portion **221** and a pair of restraining arms **222A** and **222B** that extend upward from the base portion **221**. Each of the restraining arms **222A, 222B** has a restraining tab **223A, 223B**, respectively, at the top end of the restraining arms **222A, 222B**. The restraining tabs **223A, 223B** extend inward toward each other and meet in the middle of the space between the two restraining arms **222A, 222B**.

[0084] As shown in FIGS. **6A** and **6B**, the bone staple **1000** is loaded onto the inner frame **220** so that the two legs **1011, 1012** of the staple **1000** are held in the open configuration (i.e., the two legs

substantially parallel to each other), preventing them from closing back to their relaxed position, and the bridge portion **1020** is braced against the restraining tabs **223A**, **223B**.

[0085] To help keep the two legs **1011**, **1012** of the bone staple **1000** in the open configuration, the base portion **221** of the inner frame **220** is provided with two recesses **221A** and **221B**, into which the two staple legs are inserted. This can be better seen in FIG. 6A.

[0086] The two restraining tabs **223A**, **223B** are not permanently connected to each other so the staple **1000** can be removed from the inner frame **220** by spreading the two restraining arms **222A** and **222B** apart.

[0087] The base portion **210** of the cartridge **200** comprises two guiding towers **211**, **212** that are spaced apart at a set distance for receiving and guiding the universal staple inserter **400**. FIGS. 7A-7D show the engaging action of the cartridge **200** and the universal staple inserter **400**. The universal staple inserter **400** is configured so that the staple inserter **400** and the configuration of the cartridge **200** enables transfer of the bone staple **1000** from the cartridge **200** to the universal staple inserter **400** in one single engaging motion. The engaging motion is the simple motion of inserting the universal staple inserter **400** between the two guiding towers **211**, **212** straight down as represented by the downward arrow in FIG. 7A.

[0088] The universal staple inserter **400** comprises two pairs of staple grabbers **410A**, **410B**. Each pair of staple grabbers **410A**, **410B** has two arms, one on each side of the inserter **400**, extending downward. The leading ends of the staple grabbers **410A**, **410B** have retaining tabs **411A**, **411B** that extend downward beyond the body of the universal staple inserter **400**.

[0089] As the universal staple inserter **400** is lowered into the cartridge assembly **200** between the two guiding towers **211**, **212**, the retaining tabs **411A**, **411B** of the staple grabbers **410A**, **410B** encounter the bridge portion **1020** of the bone staple **1000** causing the retaining tabs **411A**, **411B**, and in turn the staple grabbers **410A**, **410B**, to spread apart. The retaining tabs **411A**, **411B** have a wedge-like shape as shown in FIG. 7C to facilitate this action. As the universal staple inserter **400** is pushed down further, the retaining tabs **411A**, **411B** slides over the bridge portion **1020** then snaps back toward their resting position so that the arms of each pair of the staple grabbers **410A**, **410B** close together from the opposite sides with the bridge portion **1020** of the staple **1000** caught in between. At this stage, the retaining tabs **411A**, **411B** are now underneath the bridge portion **1020**. This position is shown in FIG. 7C. The wedge-like shape of the retaining tabs **411A**, **411B** helps to retain the staple **1000**.

[0090] Additionally, as the universal staple inserter **400** is being lowered into the cartridge assembly **200** and the retaining tabs **411A**, **411B** are passing by the bridge portion **1020**, the leading end of the main body of the universal staple inserter **400** gets wedged in between the two restraining tabs **223A**, **223B** of the inner frame **220** and urge them apart so that the restraining tabs **223A**, **223B** no longer are holding down the staple as the retaining tabs **411A**, **411B** of the staple grabbers **410A**, **410B** grab hold of the staple **1000**.

[0091] With the transfer of the staple **1000** from the inner frame **210** of the cartridge **200** to the universal staple inserter **400** complete, the inserter **400** can now be removed from the cartridge assembly **200** as shown in FIG. 7D. The universal staple inserter **400** is now holding the staple **1000**. The retaining tabs **411A**, **411B** of the staple grabbers **410A**, **410B** keep the bridge portion **1020** of the staple **1000**, which is normally bowed upward in the relaxed shape as shown in FIG. 5A, substantially straight as shown. This maintains the staple **1000** in its insertion shape with its two legs in substantially parallel configuration so that the staple can be inserted into a bone repair site. The universal staple inserter **400** is configured so that the two pairs of staple grabbers **410A**, **410B** can be quickly spread apart and release the staple after the staple has been inserted into a bone repair site.

[0092] The cartridge assembly **200** and the associated universal staple inserter **400** allows one staple inserter to be universally used with staples of a variety of sizes. As long as the central portions of the bridge portion **1020** of the staples of varying sizes are configured to have the same

bowed shape, the universal staple inserter **400** can be used in conjunction with the cartridge assembly **200** holding each of the staples. The cartridge assembly **200** would be provided in different sizes to hold different size staples. The two guiding towers **211**, **212** of the cartridge assemblies **200** will all be the same size to receive the universal staple holder **400**.

[0093] FIG. **8** is an illustration of an overlapping view of three different size staples **1000**, **1000C**, and **1000D** that are configured for using with the universal staple inserter **400**. The bridge portions of the three staples have a Central Portion that have the same bowed shape.

[0094] Referring to FIGS. **14A-14I**, a bone staple inserter system according to another embodiment is disclosed. The bone staple inserter system comprises a cartridge **777** shown in FIG. **14A**. A bone staple such as the two-legged inline staple **1000** shown in FIG. **5A** is securely mounted to the cartridge **777**. As described above, the bone staple **1000** comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion **1020** connecting the first and second legs, where the cartridge **777** comprises a first end and a second end, the first end provided with a channel **710** that is sized to receive the bridge portion **1020** of the bone staple **1000** and securely hold the bone staple **1000** by an interference fit.

[0095] The bone staple inserter system also comprises a bone staple inserter **700A** that is configured to receive the cartridge **777**.

[0096] Referring to FIG. **14A**, the cartridge **777** also comprises a connector part **720** that is configured to securely attach to the bone staple inserter **700A**. In the illustrated example, the connector part **720** is a cylindrical structure and comprises two tabs **722** configured for releasably attaching to the bone staple inserter **700A** by a twisting motion.

[0097] Referring to FIGS. **14B-14D**, the bone staple inserter **700A** has a body **750**. The body has a cartridge engaging end **751** that is configured with a recess **720A** for receiving the connector part **720** of the cartridge. Around the periphery of the recess **720A** is provided with two L-shaped cutouts **722A** for receiving the two tabs **722** on the connector part **720** of the cartridge **777**. The L-shape of the cutouts **722A** allows the tabs **722** to first engage the cartridge engaging end **751** straight then twisted to temporarily lock the cartridge **777** with the inserter **700A**.

[0098] The inserter **700A** further comprises a pushrod **770** that protrudes through the recess **720A** for engaging the cartridge **777** and release the staple **1000** from the cartridge **777**. The pushrod **770** is configured to move in and out of the cartridge engaging end **751** as indicated by the arrow A in FIG. **14B**. The movement of the pushrod **770** is controlled by a lever **760**. The lever **760** is connected to the pushrod **770** on the opposite end of the pushrod **770** by a hinged connection so that when the lever **760** is pulled up as shown in FIG. **14B**, the pushrod **770** is in its retracted position. When the lever **760** is pushed down into the body **750** as shown in FIG. **14C**, the pushrod **770** is in its fully extended position.

[0099] An exemplary mechanical arrangement that enables the reciprocating motion of the pushrod **770** by the manipulation of the lever **760** is shown in the cross-sectional view of the staple inserter **700A** shown in FIG. **14G**. In this example, the pushrod **770** is connected to the lever **760** via an intermediate linkage **770A**. The connections between the intermediate linkage **770A** and the pushrod **770** and the lever **760** are both hinged connections to accommodate the angular bending at those connection points when the lever **760** goes from the pushed-down position of FIG. **14G** to the pulled-up position of FIG. **14B**. The lever **760** is connected to the body **750** by a hinge pin **765** as shown in the side view of FIG. **14H**. The hinge pin **765** defines the fulcrum F (shown in FIG. **14G**) and provides the hinged movement for the lever **760**. The hinged connection between the intermediate linkage **770A** and the lever **760** is identified by the point H in FIG. **14G**. Because the hinge point H is between the pushrod **770** and the fulcrum F, when the lever **760** is pulled up, the hinge point H rises with the lever **760** and the intermediate linkage **770A** pulls the pushrod **770** into the body **750** into its retracted position. When the lever **760** is pushed back down, the reverse happens and the intermediate linkage **770A** pushes the pushrod **770** out to its extended position.

[0100] Referring to FIG. **14E** which shows the empty cartridge **700B** attached to the inserter **700A**,

the pushrod **770** in its fully extended position can be seen. In this position, the pushrod protrudes into the channel **710**. Thus, when a bone staple **1000** is securely loaded into the channel **710** by a press-fitting or an interference-fitting, the bowed portion of the staple's bridge **1020** sits within the channel **710**. In use, the lever **760** on the staple inserter **700A** is first pulled up to full retract the pushrod **770**. Then, the connector part **720** of a loaded cartridge **700B** is inserted into the recess **720A** of the staple inserter **700A** and twisted to lock the engagement. At this point, the staple **1000** is in its relaxed shape and the two legs of the bone staple **1000** are biased toward each other as shown in FIG. **14A**. Next, the staple **1000** is changed into its insertion shape by pushing the lever **760** down to the position shown in FIGS. **14C** and **14E**. This extends the pushrod **770** into the channel **710** which pushes on the bridge **1020** of the staple **1000** from the convex side of the bowed bridge **1020**. This straightens the bridge **1020** causing the two legs of the staple **1000** to be substantially parallel to each other. With the staple **1000** engaged with the inserter **700A** in this manner and the lever **760** is in the down position, the angle of the intermediate linkage **770A** creates an over center connection where the staple pushing back on the pushrod **770** keeps the linkage formed by the pushrod **770**, the intermediate linkage **770A**, and the lever **760** rigid (similar to a person locking a straightened elbow) and keeps the system engaged and prevents the lever from popping up to the pulled up position when not desired. Now, the staple **1000** is in its insertion shape and the surgeon can insert the staple **1000** into the pre-drilled holes in the bone repair site. After the insertion, the inserter **700A**/cartridge **700B** assembly can be removed by pulling up on the lever **760** to retract the pushrod **770** and relieve the pressure on the bridge **1020** returning the staple **1000** to its relaxed shape. Because the staple **1000** is now held only by the friction between the channel **710** in the cartridge **700B** and the bridge **1020**, the inserter **700A**/cartridge **700B** assembly can be removed by laterally twisting the assembly.

[0101] Referring to FIG. **9**, according to an aspect of the present disclosure, a bone staple inserter kit **500** for implanting two-legged inline bone staples such as those illustrated in FIGS. **5A-5E** is disclosed. The bone staple inserter kit **500** comprises a first sterile package **500A** containing a first bone staple securely mounted to a first cartridge assembly **200** described above. The first bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs.

[0102] The bone staple inserter kit **500** also comprises second sterile package **500B** containing a bone staple inserter **400** described above that is configured to receive the cartridge assembly **200** loaded with a bone staple. The second sterile package **500B** also contains a universal drill guide assembly (like the drill guide assemblies **100**, **100A**, or **300**). The second sterile package **500B** can also contain other hardware components **10** necessary to use the drill guide. Some examples are one or more drill bits, fixation pins, etc.

[0103] In some embodiments, the bone staple inserter kit **500** can further comprise one or more additional sterile packages, like the sterile package **500A**, each containing an additional cartridge assembly securely holding a bone staple, where the bone staples in the additional sterile packages can be of same or difference size as the bone staple in the first sterile package.

[0104] By providing bone staple inserter kits such as **500** described herein, only one set of universal instruments need to be provided in one sterile package **500B** and several different sterile packages **500A** each containing a different size staple can be provided.

[0105] Referring to FIG. **15**, according to an aspect of the present disclosure, a bone staple inserter kit **800** is disclosed. Such kit comprises a first sterile package **801A** containing a cartridge assembly **777** and a second sterile package **802** containing a set of universal instruments needed for implanting bone staples.

[0106] The cartridge assembly **777** provided in the first sterile package **801A** comprises a bone staple securely mounted to the first cartridge as described herein with reference to FIG. **14A**. The universal instruments provided in the second sterile package **802** can comprise the staple inserter **700A** described with reference to FIG. **14B**. The second sterile package **802** preferably also

comprises one or more of the universal drill guide **100**, **100A**, and **300** described herein.

[0107] In some embodiments, the bone staple inserter kit **800** can further comprise one or more additional first sterile packages **801B**, **801C**, etc. Each of these additional first sterile packages can contain a cartridge assembly holding a staple of different size or a same size as the staple in the first sterile package **801A**.

[0108] Although the devices, kits, systems, and methods have been described in terms of exemplary embodiments, they are not limited thereto. Rather, the appended claims should be construed broadly, to include other variants and embodiments of the devices, kits, systems, and methods, which may be made by those skilled in the art without departing from the scope and range of equivalents of the devices, kits, systems, and methods.

## Claims

1. A bone staple comprising: a first leg having a wedge-shaped tip; a second leg having a wedge-shaped tip, wherein each of the first and second legs further includes an inner surface each having a plurality of barbs, each barb having a tip that is located flush with the inner surface of the leg; and a bridge extending between a first end of the first leg and a second end of the second leg, the bridge comprising a top surface along the length of the bridge, wherein the top surface comprises two chamfered edge surfaces that extend substantially the length of the bridge and one flat surface located between the two chamfered edge surfaces, wherein the flat surface extends substantially the length of the bridge.
2. The bone staple of claim 1, the bridge forms a first transition to the first leg at the first end and a second transition to the second leg at the second end wherein each of the first end transition and the second end transition has a first radius of curvature  $R1$  and a transitions from the bridge to the legs along a bottom surface of the bridge has a second radius of curvature  $R2$ , wherein  $R1 > R2$ .
3. The bone staple of claim 1, wherein the staple is comprised of a shape memory material such that the staple is movable between an insertion shape and a relaxed shape.
4. A bone staple comprising: a pair of spaced-apart legs, each having a proximal end and a distal end, the pair of spaced-apart legs defining a compression plane; a non-linearly extending bridge connecting the distal ends of the pair of spaced-apart legs, wherein the bridge has a generally U-shape that comprises: a flat portion that spans the distance between the distal ends of the pair of spaced-apart legs; and two ends of the U-shape connecting the flat portion to the proximal ends of the spaced-apart legs, wherein the two ends of the U-shape extending laterally between the flat portion and the proximal ends of the spaced-apart legs, whereby the flat portion is off-set from the compression plane, wherein the flat portion has a width that is greater than its thickness.
5. The bone staple of claim 4, wherein each of the pair of spaced-apart legs comprises a surface arranged confronting spaced-apart relation to one another each having a plurality of barbs along the surface.
6. The bone staple of claim 5, wherein the staple is comprised of a shape memory material such that the staple is movable between an insertion shape and a relaxed shape.
7. A bone staple inserter kit comprising: a first sterile package containing a first bone staple securely mounted to a first cartridge, wherein the first bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs, wherein the first cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge portion of the first bone staple and securely hold the first bone staple by an interference fit; and a second sterile package containing; a bone staple insertion tool that is configured to receive a cartridge, like the first cartridge that has a bone staple, like the first bone staple, securely mounted thereon; a universal drill guide assembly; and a drill bit, wherein the bone staple insertion tool comprises an actuation mechanism that can transform the bone staple to an insertion shape after the cartridge with the bone staple mounted

thereon is mounted on the insertion tool.

- 8.** The bone staple inserter kit of claim 7, wherein the actuation mechanism applies pressure on the bridge portion of the staple to straighten the curved bridge portion after the cartridge that has the bone staple is mounted on the insertion tool, wherein straightening the curved bridge portion expands the first and second legs of the bone staple into the insertion shape.
- 9.** The bone staple inserter kit of claim 7, further comprising a third sterile package containing a second bone staple securely mounted to a second cartridge, wherein the second bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs, wherein the second cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge portion of the second bone staple and securely hold the second bone staple by an interference fit, wherein the first bone staple and the second bone staple have difference sizes.
- 10.** A surgical staple comprising: a bridge having an elongated shape with a longitudinal axis defined through its length, the bridge further comprising first, second, third, and fourth corners; a first leg extending from the first corner; a second leg extending from the second corner; a third leg extending from the third corner; and a fourth leg extending from the fourth corner; wherein the bridge comprises a first section that includes the first and second corners, and a second section that includes the third and fourth corners; and wherein the first and second corners are arranged such that a line connecting the first and second corners intersect the longitudinal axis of the bridge at a first non-orthogonal angle.
- 11.** The surgical staple of claim 10, further wherein the third and fourth corners are arranged such that a line connecting the third and fourth corners intersect the longitudinal axis of the bridge at a second non-orthogonal angle.
- 12.** The surgical staple of claim 11, wherein the first non-orthogonal angle and the second non-orthogonal angle are the same angle.
- 13.** The surgical staple of claim 10, wherein the line connecting the first and second corners and the line connecting the third and fourth corners are parallel.
- 14.** The surgical staple of claim 10, wherein the bridge further comprising a central reinforcement ridge that is thicker than the rest of the bridge and extending along the longitudinal axis in a portion of the length of the bridge.
- 15.** The surgical staple of claim 10, wherein each of the legs have a length that is independent of one another.
- 16.** The surgical staple of claim 10, wherein the legs are of the same length.
- 17.** The surgical staple of claim 10, wherein the bridge has an upper surface that curves down ward from the center line to the four corners.
- 18.** A system comprising; a bone staple including a pair of spaced-apart legs, each having a proximal end and a distal end, the pair of spaced-apart legs defining a compression plane; a bridge connecting the distal ends of the pair of spaced-apart legs, wherein the bridge has a generally curved shape with a flat portion that spans the distance between the distal ends of the pair of spaced-apart legs and two ends coupling the flat portion to the proximal ends of the spaced-apart legs, wherein the two ends extend laterally between the flat portion and the proximal ends of the spaced-apart legs; a first sterile package containing the bone staple securely mounted to a first cartridge, wherein the pair of spaced-apart legs are oriented toward each other in a relaxed state, wherein the first cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge of the bone staple and securely hold the bone staple by an interference fit; and a second sterile package containing a bone staple insertion tool that is configured to receive the cartridge so as to secure the bone staple for implantation in a patient wherein securing the bone staple for implantation by the bone staple insertion tool comprises transforming the bone staple to a biased insertion shape after the cartridge with the bone staple mounted thereon is mounted on the insertion tool.

**19.** The system for delivering a bone staple to a surgical theater of claim 18, wherein transforming the bone staple to a biased insertion shape includes applying pressure on the bridge of the bone staple to straighten the bridge after the cartridge that has the bone staple is mounted on the insertion tool, wherein straightening the bridge expands and thereby biases the first and second legs of the bone staple into the insertion shape.

**20.** The system for delivering a bone staple to a surgical theater of claim 18, wherein, further comprising a third sterile package containing a second bone staple securely mounted to a second cartridge, wherein the second bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs, wherein the second cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge portion of the second bone staple and securely hold the second bone staple by an interference fit, wherein the first bone staple and the second bone staple have difference sizes.

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